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EXPERIMENT: LENGTH MEASURING INSTRUMENTS

Objectives :

1. Measuring the quantity of length, especially the diameter of the object, with a length measuring instrument.
2. Calculate the uncertainty value of the measurement results of the length measuring instrument.

EXPERIMENT SUMMARY

INTRODUCTION

Science and engineering are essentially based on measurement and comparison [1]. Measuring means comparing a physical quantity with a standard that has been set and agreed upon internationally as a reference unit. The International System (SI), established at the 14th General Conference on Weights and Measures in 1971, defines the basic units of measurement, namely meters for length, kilograms for mass, and seconds for time [1]. Without a uniform and standardized system of units, there is no basis for comparing measurement results from one laboratory to another, let alone between countries.

Length measurement activities are often found in various professional fields. However, what distinguishes scientific measurement from everyday measurement is the demand for precision (the level of consistency and repeatability of results when measurements are made many times) as well as explicit reporting of uncertainties. No single measurement can be made with absolute precision. Every measurement must contain uncertainty (*uncertainty*), which comes from the limitations of the device's resolution (the ability of the device to detect the smallest possible size change), the ability to read the observer, and the variation in conditions during the measurement [2]. Therefore, stating the measurement results without the uncertainty value is incomplete reporting and cannot be scientifically accounted for. The uncertainty value of a measurement is influenced by the quality of the instrument, the skill of the observer, and the number of repetitions performed [3]. The more repetition and the better the instrument used, the less uncertainty attached to the measurement results.

In this practicum, three length measuring instruments with different levels of accuracy were used, namely ruler, caliper, and Micrometer. Each tool has the smallest scale value (NST), which is the smallest measurement value that can still be read and measured with certainty by the ruler used has an NST of 1 mm, the caliper used has an NST of 0.02 mm, and the micrometer has an NST of 0.01 mm. Therefore, the micrometer is the most meticulous instrument among the three [3]. The selection of instruments that do not match the scale of the object being measured will result in data with disproportionate uncertainty, thereby reducing the validity of the data.

Data collection in the experiment can be done singly or repeatedly. For a single measurement on a linear scale tool such as a bar, the observer can make a visual estimate if the boundary of the object is between two scale lines. Visual estimation is also possible on the micrometer rotary scale because the distance between the measurement lines still allows the observation of the median estimate. In contrast, in the caliph, the data reading is based on the principle of coincident, i.e. the squeeze of a single line between the main scale and the vernier scale that forms a single straight line so that visual estimates are difficult to make and intermediate estimates are not required. Regardless of the reading method, the uncertainty (Δx) for a single measurement on an analog device is generally defined as $\Delta x = \text{NST}/2$ [4].

The analysis of repeated measurement data is carried out through three main statistical quantities. First, the standard deviation that represents the degree of distribution or variation of a set of measurement data. Standard deviation can be formulated in the following equation:

$$\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}} \quad (1)$$

The parameter x_i is the value of the data to- i , $\bar{x}$ is the average value of the entire data, and n is the number of measurements taken [4]. A high standard deviation value indicates high data variation and vice versa. Second, the standard deviation from the mean (Δx), which represents the absolute uncertainty of the average value of the measurement results, is formulated as follows:

$$\Delta x = \frac{\sigma}{\sqrt{n}} \quad (2)$$

A low standard deviation value of the mean (Δx) can be interpreted as an indication that the data has a high precision to the average of the data sample. Third, the Δx relative error (I) which expresses uncertainty relative to the quantity measured in percent whose equation is as follows [4]:

$$I = \frac{\Delta x}{\bar{x}} \times 100\% \quad (3)$$

The formulation shows that the smaller the percentage, the more precise the data obtained is to the average value. Furthermore, the final measurement results are expressed in the following standard notation:

$$x = \bar{x} \pm \Delta x \quad (4)$$

which means that the value of the actual quantity lies between $\bar{x} - \Delta x$ and $\bar{x} + \Delta x$ [4], [5].

METHODOLOGY

A. Tools and Materials

The following are the tools and materials used in this practicum:

Table 1. Tools and materials for the long measuring tool

Apparatus	Quantity	Function
Ruler	1 piece	Used to measure the diameter of marbles
Micrometer	1 piece	Used to measure the diameter of marbles
Caliper	1 piece	Used to measure the diameter of marbles
Large marbles	1 piece	Measured objects
Small marbles	1 piece	Measured objects

B. Procedure

1. Caliper

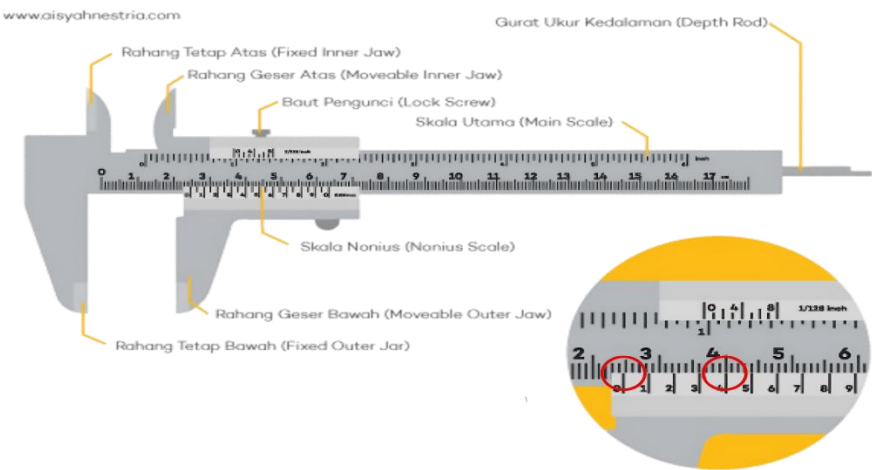


Figure 1. Caliper.
Source: aisyahnestria.com

Because it is difficult to visually determine the median estimate (the approximate value between the two closest scales), the uncertainty of a single measurement using a caliper is taken as half of the value of the smallest scale (NST) of the tool. With a vernier scale of 0.02 mm, the uncertainty of a single measurement is:

$$\Delta x = \frac{1}{2} \cdot 0,02 \text{ mm} = 0,01 \text{ mm}$$

The way to determine the value of x (the length of the object) is as follows:

- Note the number on the main scale that is adjacent to the number 0 on the vernier scale. In Figure 1 the figure is 2.2 cm.
- Note the vernier line that is conjoined by the main scale. In Figure 1, the line that is conjoined is the fourth line, so the value is $x = 24,00 \text{ mm} + (4 \times 0,1 \text{ mm}) = 24,40 \text{ mm}$. Thus, the length measurement results are written as $24,40 \pm 0,01$

The working steps of measuring using caliper are as follows:

1. The diameter of small and large marbles is measured using a caliper.
2. Measurements were taken by five different people in turn.
3. Each person takes one measurement so that 5 repetition data is obtained.
4. All measurement data are recorded in the available data table.
5. Steps 1–4 are repeated using another measuring instrument.

2. Micrometer

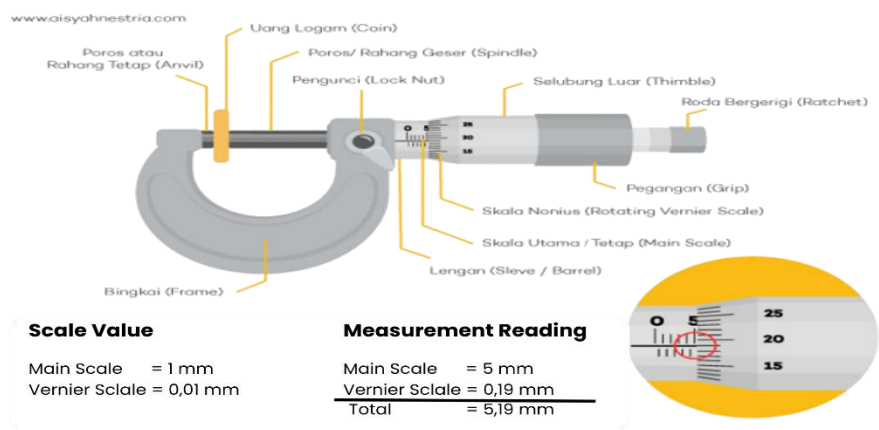


Figure 2. Micrometer.
Source: aisyahnestria.com

When the micrometer vernier scale is rotated one full turn, the sliding jaw and the vernier scale will move forward or backward by 0.5 mm. Since the vernier scale is divided into 50 scales, then the micrometer NST is 0.5 mm / 50 = 0.01 mm. Thus, the uncertainty of the single measurement is:

$$\Delta x = \frac{1}{2} \cdot 0,01 \text{ mm} = 0,005 \text{ mm}$$

The way to determine the value of *x* (the length of the object) is as follows:

- Notice the position of the main scale line to the vernier scale. In Figure 2, the main scale shows the figure of 5 mm.
- It is observed that the horizontal line on the vernier scale is conjoined with the horizontal line on the main scale. In Figure 2, the line that is conjoined is at 19, so the value of *x* = 5 + (19 × 0,01) = 5,19 mm. Thus, the length measurement results are written as 5,19 ± 0,005 mm.

The working steps of measuring using a micrometer are as follows:

1. The diameter of small and large marbles is measured using a micrometer.
2. Measurements were taken by five different people in turn.
3. Each person takes one measurement so that 5 repetition data is obtained.
4. All measurement data are recorded in the available data table.
5. Steps 1–4 are repeated using another measuring instrument.

3. Ruler

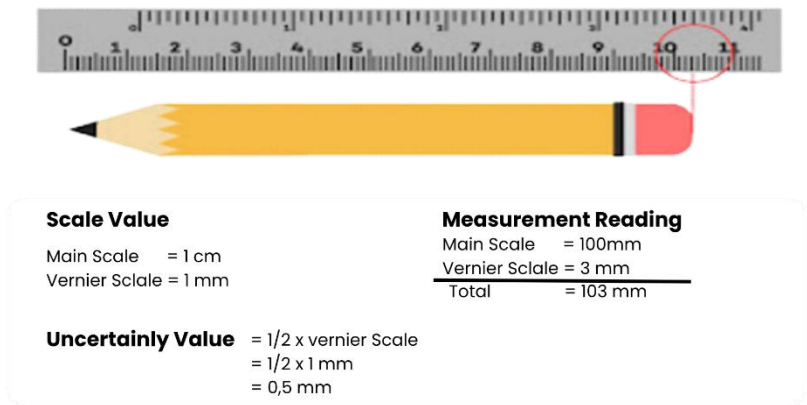


Figure 3. Ruler
Source: aisyahnestria.com

The working steps of measuring using a bar are as follows:

1. The diameter of small and large marbles is measured using a bar.
2. Measurements were taken by five different people in turn.
3. Each person takes one measurement so that 5 repetition data is obtained.
4. All measurement data are recorded in the available data table.
5. Steps 1–4 are repeated using another measuring instrument.

D. Experiment Flowchart

DATA ANALYSIS AND DISCUSSION

A. Analysis of Measurement Data

The following is a table of data from the results of measurements made in the laboratory:

Table 2. The measurement results of small marble diameters.

Repetition to-	Caliper *(±..... mm)	Micrometer *(±..... mm)	Ruler *(±..... mm)
1			
2			
3			
4			
5			

*Filled with uncertainty of measuring instruments

Table 3. The measurement results of large marble diameters.

Repetition to-	Caliper *(±..... mm)	Micrometer *(±..... mm)	Ruler *(±..... mm)
1			
2			
3			
vern			
5			

*Filled with uncertainty of measuring instruments

B. Calculation Analysis

Example calculation: (average, Standard deviation, Standard average deviation, relative error, measurement result)

The following is a table of the data obtained from the calculations:

Table 4. The result of the calculation of the diameter of the marbles is small.

Repetition to-	Caliper *(±..... mm)	Micrometer *(±..... mm)	Ruler *(±..... mm)
1			
2			
3			
4			
5			
Average			
Standard deviation			
Standard average deviation			
Relative error			
Measurement results			

Table 5. The results of the calculation of the diameter of large marbles.

Repetition to-	Caliper *(±..... mm)	Micrometer *(±..... mm)	Ruler *(±..... mm)
1			
2			
3			
4			
5			
Average			
Standard deviation			
Standard average deviation			
Relative error			
Measurement results			

C. Discussion

CONCLUSION

From this practicum it can be concluded:

- 1.

REFERENCES

- [1] D. Halliday, R. Resnick, and J. Walker, *Fundamentals of Physics Extended*, 9th ed. Hoboken, NJ: John Wiley & Sons, 2010.
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- [4] J. R. Taylor, *An Introduction to Error Analysis: The Study of Uncertainties in Physical Measurements*, 3rd ed. Sausalito, CA, USA: University Science Books, 2022.
- [5] P. R. Bevington and D. K. Robinson, *Data Reduction and Error Analysis for the Physical Sciences*, 3rd ed. New York: McGraw-Hill, 2003.

PRELIMINARY ASSIGNMENT

1. Mention and explain the difference between precision and accuracy in physics measurements.
2. An object was measured 5 times in length by different people and obtained the following data: 12.30 cm; 12.35 cm; 12.28 cm; 12.33 cm; and 12.29 cm. Determine:
 - a. Average value of measurement results
 - b. Standard deviation
 - c. Standard deviation from the mean
 - d. Relative error
 - e. Write down the measurement results in the form of $x = \bar{x} \pm \Delta x$